TongueReenact: Geometry-Anchored Tongue Synthesis for Face Reenactment
Supplementary Material

MD Wahiduzzaman Khan[1], Mingshan Jia[1], Xiaolin Zhang[2⋆], En Yu[1], and Katarzyna Musial-Gabrys[1]

[1]University of Technology Sydney, Australia

arnobk511@gmail.com, mingshan.jia@uts.edu.au,

en.yu-1@uts.edu.au, musial.katarzyna@gmail.com

[2]Shandong University of Science and Technology, China

solli.zhang@gmail.com

A. Extended Tongue Segmentation Evaluation

We report comprehensive segmentation metrics comparing TongueSAM [1], our initial seed model, and our final bootstrapped BiSeNet* across 6,344 evaluation frames from NERSemble [3]. Tables S1 and S2 report performance on tongue-present frames. Table S3 reports hallucination behavior on tongue-absent frames.

TongueSAM achieves near-zero precision, producing predicted masks 10–115× larger than ground truth depending on the exposure group. While TongueSAM achieves the highest recall, this is a consequence of extreme over-segmentation rather than accurate detection, as confirmed by area ratios of 10–115× ground truth. Our bootstrapped BiSeNet* trades recall for substantially higher precision and IoU, yielding masks that are geometrically accurate rather than over-inclusive. Our initial model substantially improves localization across all groups but overestimates mask area in the Medium group (ratio 3.07), reflecting domain shift on partially visible tongue frames. The final bootstrapped BiSeNet* achieves the best IoU and precision across all groups, with area ratios closest to 1.0, and suppresses hallucination on tongue-absent frames to 18%, compared to 100% for both TongueSAM and our initial model.

Table S1. Overall segmentation metrics on tongue-present frames ($N = 3{,}131$).

Metric	TongueSAM	Our Initial	Bootstrapped
IoU ↑	0.104	0.597	**0.640**
Precision ↑	0.105	0.653	**0.783**
Recall ↑	**0.895**	0.872	0.727
F1 ↑	0.156	0.711	**0.730**
Area Ratio → 1	54.46	1.842	**0.979**

⋆ Corresponding author.

Table S2. Per-group segmentation metrics on tongue-present frames.

Group	Metric	TongueSAM	Our Initial	Bootstrapped
Small ($N = 959$)	IoU ↑	0.103	0.724	**0.752**
	Precision ↑	0.104	0.784	**0.866**
	Recall ↑	0.889	**0.902**	0.843
	F1 ↑	0.159	0.828	**0.842**
	Area Ratio → 1	28.42	1.216	**1.015**
Medium ($N = 1{,}148$)	IoU ↑	0.010	0.345	**0.376**
	Precision ↑	0.010	0.367	**0.599**
	Recall ↑	**0.855**	0.848	0.474
	F1 ↑	0.021	**0.488**	0.484
	Area Ratio → 1	115.15	3.068	**0.899**
Large ($N = 1{,}024$)	IoU ↑	0.210	0.760	**0.832**
	Precision ↑	0.211	0.852	**0.911**
	Recall ↑	**0.945**	0.870	0.902
	F1 ↑	0.304	0.851	**0.901**
	Area Ratio → 1	10.81	1.052	**1.034**

Table S3. Mean predicted mask area (pixels) on tongue-absent frames. Lower values indicate fewer false positives. Group-level counts are small ($N = 12$) and should be interpreted accordingly.

Group	TongueSAM	Our Initial	Bootstrapped
Medium ($N = 12$)	35,470	1,500	**0.42**
Large ($N = 12$)	65,606	2,241	**85.2**
Overall ($N = 3{,}213$)	56,189	1,577	**343**

B. VLM-Based Evaluation Details

B.1. Annotation Protocol

Expert annotators evaluated 100 frames across four ablation variants: Full Model, w/o Tongue, w/o Dilation, and w/o Reenactment. For each frame, annotators were presented with the source image, the driving image, and the four generated outputs side by side. Annotations covered four criteria:

- **Tongue Quality.** Rated on a 0–5 scale, where 0 indicates no visible tongue.
- **Tongue Positioning.** A binary judgement of whether the tongue is correctly positioned relative to the driving frame.

- **Best Tongue.** Which variant produces the best tongue result.
- **Best Overall.** Which variant produces the best overall reenactment.

B.2. Fine-Tuning Setup

We fine-tune Qwen3-VL-8B-Instruct [4] on the 100 annotated frames using LoRA [2]. The annotated set is split 80/20 into training and validation. Each training sample consists of six images: the source identity, the driving frame, and the four generated outputs, paired with the structured annotation as the target response.

LoRA is applied with rank $r = 64$, scaling factor $\alpha = 128$, and dropout 0.05. The multimodal LLM head is trained jointly with the LoRA adapters. Training runs for 10 epochs with an effective batch size of 16 (per-device batch size 2, gradient accumulation 8), learning rate 2×10^{-4} with a cosine schedule and warmup ratio 0.03, weight decay 0.01, and maximum sequence length 2,048 tokens, using bf16 precision on a single NVIDIA A40 GPU.

B.3. Inference Procedure

The fine-tuned model is applied to the remaining 1,000 frames using greedy decoding with a maximum of 512 new tokens. Each inference call receives the same six-image prompt as training. The model outputs a structured JSON response containing tongue quality ratings, positioning judgements, best tongue, and best overall selections for all four variants. Outputs are parsed directly from the JSON response; malformed outputs are flagged and excluded from aggregation.

C. Qualitative Video Inference Results

Figure S1 shows consecutive frame strips from three source-driving pairs. Our method transfers tongue dynamics smoothly across frames while preserving the source identity's facial structure and head pose throughout each sequence.

References

[1] Shan Cao, Qingfeng Wu, and Linjian Ma. Tonguesam: An universal tongue segmentation model based on sam with zero-shot. In *2023 IEEE international conference on bioinformatics and biomedicine (BIBM)*, pages 4520–4526. IEEE, 2023.

[2] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Liang Wang, Weizhu Chen, et al. Lora: Low-rank adaptation of large language models. *Iclr*, 1 (2):3, 2022.

[3] Tobias Kirschstein, Shenhan Qian, Simon Giebenhain, Tim Walter, and Matthias Nießner. Nersemble: Multi-view radiance field reconstruction of human heads. *ACM Transactions on Graphics (TOG)*, 42(4):1–14, 2023.

[4] An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.



Figure S1. Consecutive frame strips from three source-driving pairs. For each pair, the driving sequence (top) and our generated output on the source identity (bottom) are shown across eight frames. Our method produces temporally smooth tongue dynamics that faithfully follow the driving identity's articulation while preserving the source identity across all non-synthesized facial regions.